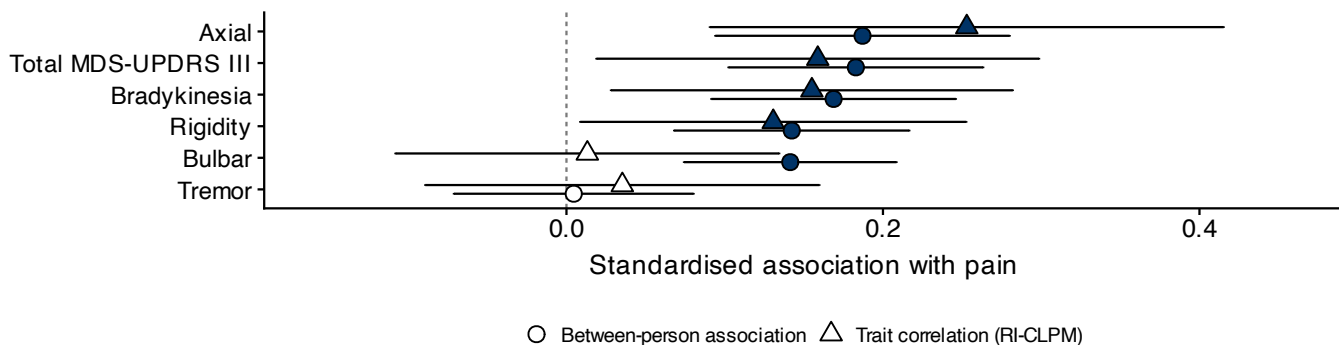
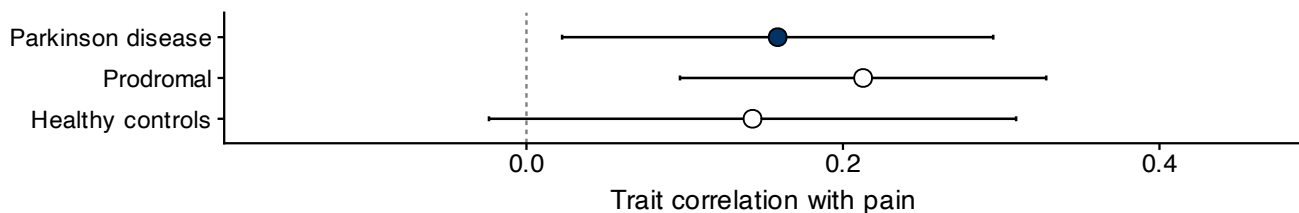


Figure 9. What the covariation is made of, and what it is not specific to

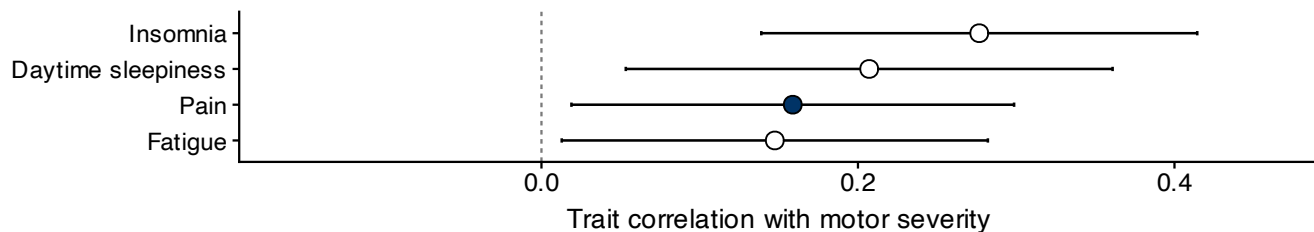
A. Which motor domain carries it



B. Which cohort it occurs in



C. Which non-motor symptom it involves



Navy fill marks an estimate whose own p value is below 0.05; open markers do not clear it. In panels B and C the open markers are also the comparators that decide whether the navy estimate is distinctive. A. The five domains sum exactly to the total. Circles are the between-person regression coefficient and triangles the latent trait correlation; these are different parameters and are shown together because they disagree in one informative place. Both put the covariation in the rigidity, bradykinesia and axial domains and neither finds it in tremor, a contrast of 0.172 within the same patients and the same instrument ($p = 0.002$). They diverge on the bulbar domain, which the regression ranks with the others and the trait model puts at zero, as a general severity factor rather than anything specific would predict. Open markers are estimates whose own p value exceeds 0.05. B. The same model refitted in the prodromal and healthy control cohorts gives correlations the data cannot distinguish from the patient estimate (Wald test of equality across the three cohorts, $p = 0.744$). This does not demonstrate equivalence; with 307 controls the interval admits both no correlation and a larger one than in patients. What it does prevent is a claim of specificity. C. Pain does not stand out among non-motor items measured the same way; insomnia covaries more strongly. Intervals in panels A and C are derived from the reported p values, since the source tables store the correlation and its p value but not its standard error.